Tutorial 1

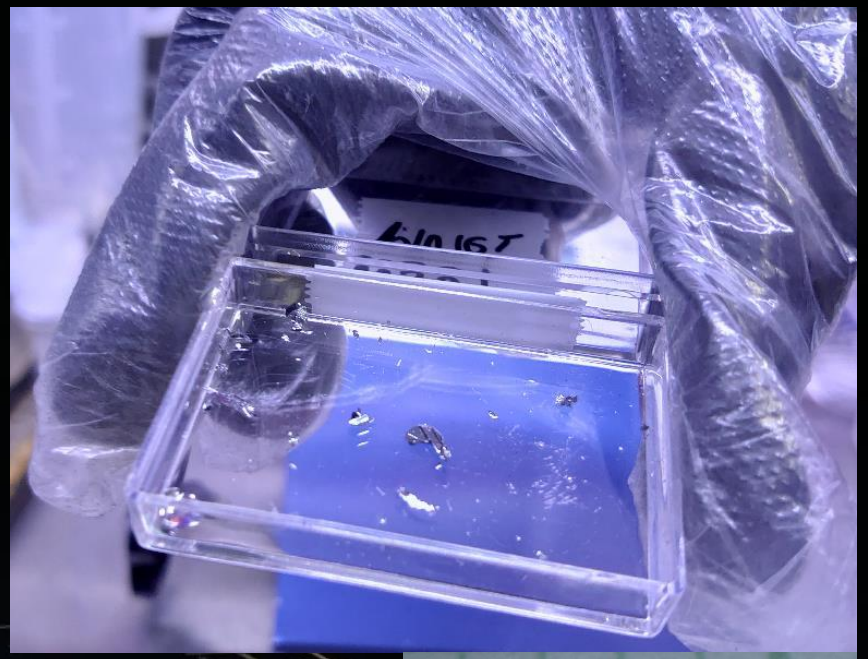
Toby Lai

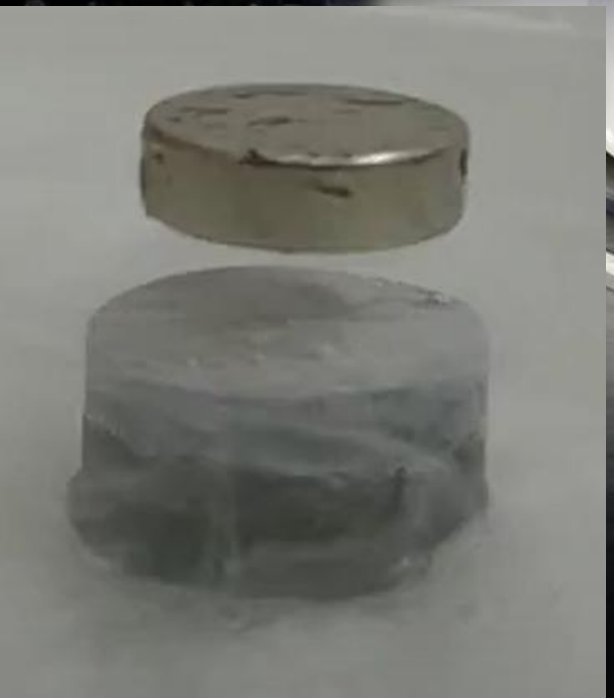
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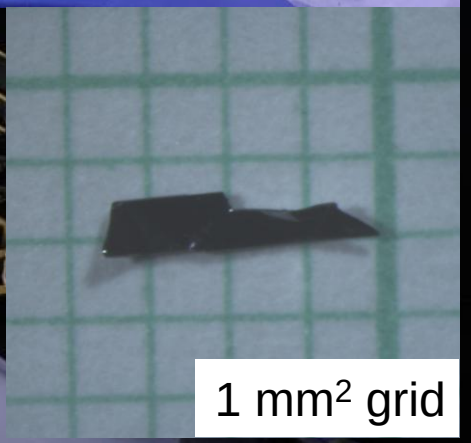
Mathematics: A Cultural Heritage

Your tutorial's teacher:
Toby Lai

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- Tel: 3917 4594
- Office: HOC 223







1 mm^2 grid

Icebreaker

- Please share…
  - YOUR NAME, and your faculty/department
- And pick one to share:
  - **1: Your Favourite Hobby**
  - **2: A Recent Happy Memory**
  - **3: Your Dream Job**
  - **4: Your Favourite Place on Earth**
  - **5: Anything About Yourself**



Expectations

- Tutorial Participation (25% of final grade)
  - You are encouraged to ask questions.
  - You will work with one another.
  - You will have many opportunities to participate.
- To respect your fellow classmates, please show up on time.
- If you can't make the class:
  - If you are sick, please send/scan me a doctor's note.
  - If you have scheduling conflict, please let me know ahead of time (if possible).



Learning Outcomes

- By the end of this tutorial you should be able to…

  Represent our Arabic numerals in the Egyptian number system and vice versa.

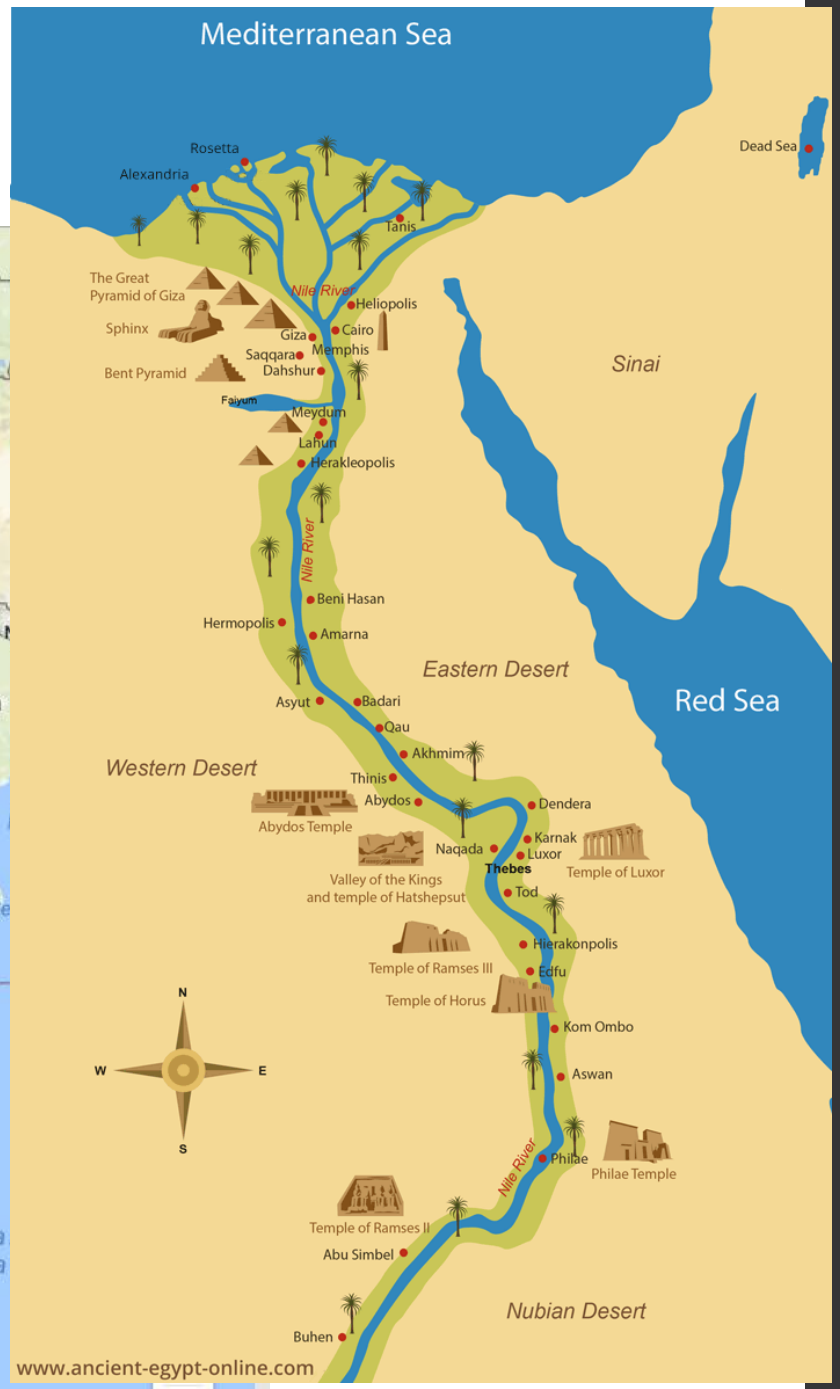
  Describe the properties of Egyptian fractions.

  Compare the understanding of mathematics in Egyptian culture with our daily understanding of mathematics.

  Provide examples of how calculations in Egyptian mathematics were fundamentally additive.

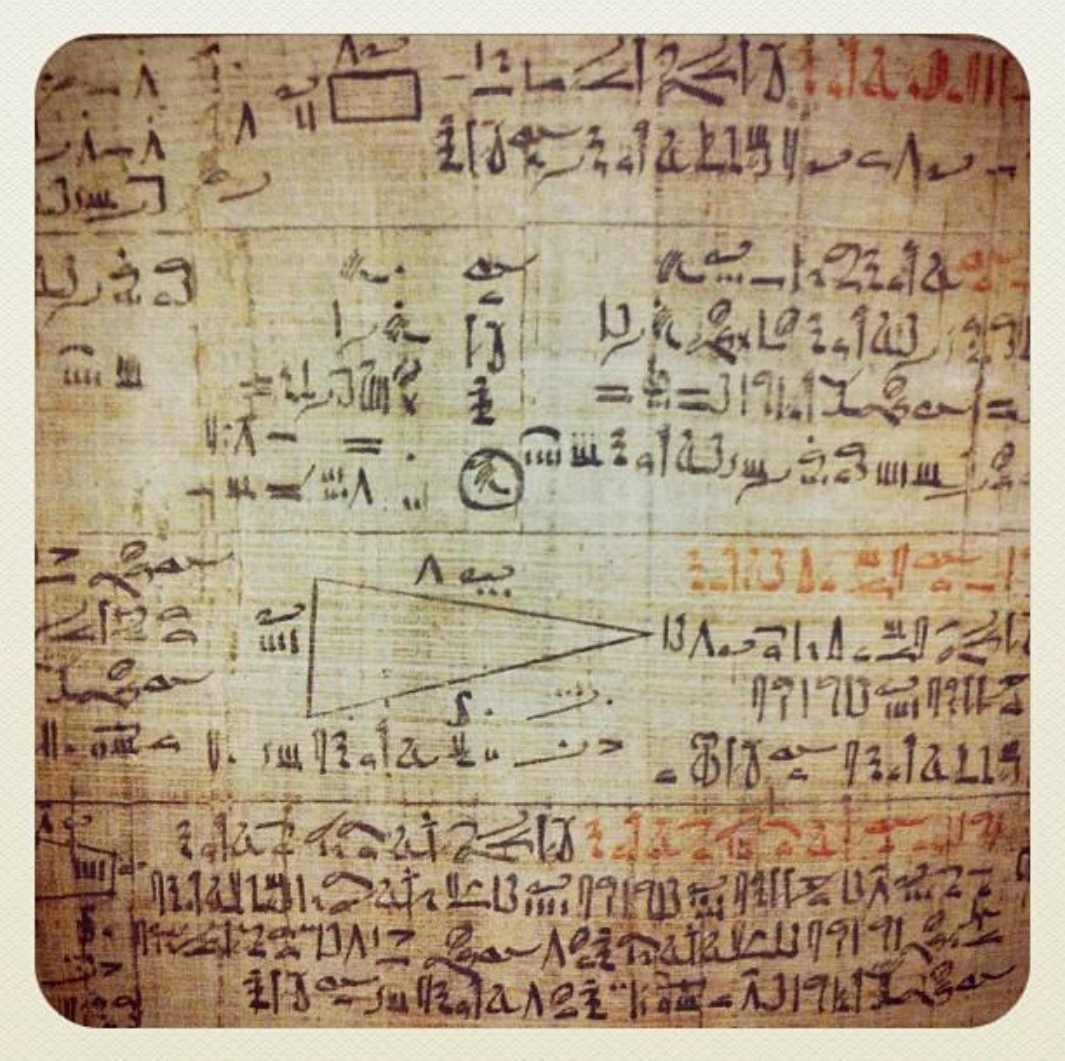
Ancient Egypt





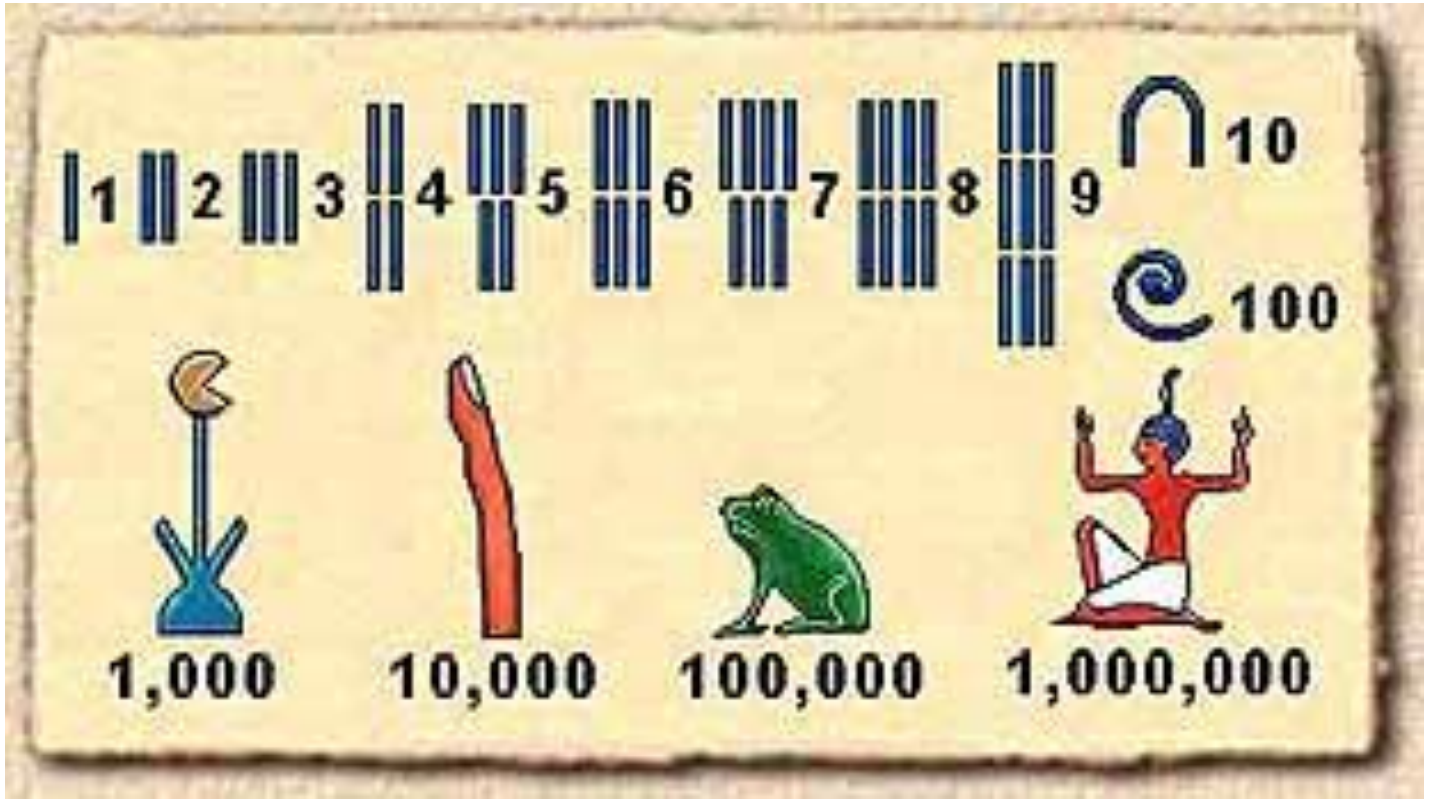
Rhind Papyrus

- Named after Scottish antiquarian: Alexander Henry Rhind (but transcribed by Ahmes)
- Stored mostly in the British museum
  - Small fragments in the Brooklyn Museum
- Book 1: Arithmetic and Algebra
  - Egyptian Numbers
  - Linear Equations
  - Dimensional analysis
- Book 2: Geometry
  - Volumes
  - Fractional equalities
- Book 3: Miscellany
  - Different miscellaneous properties based on already discussed concepts.

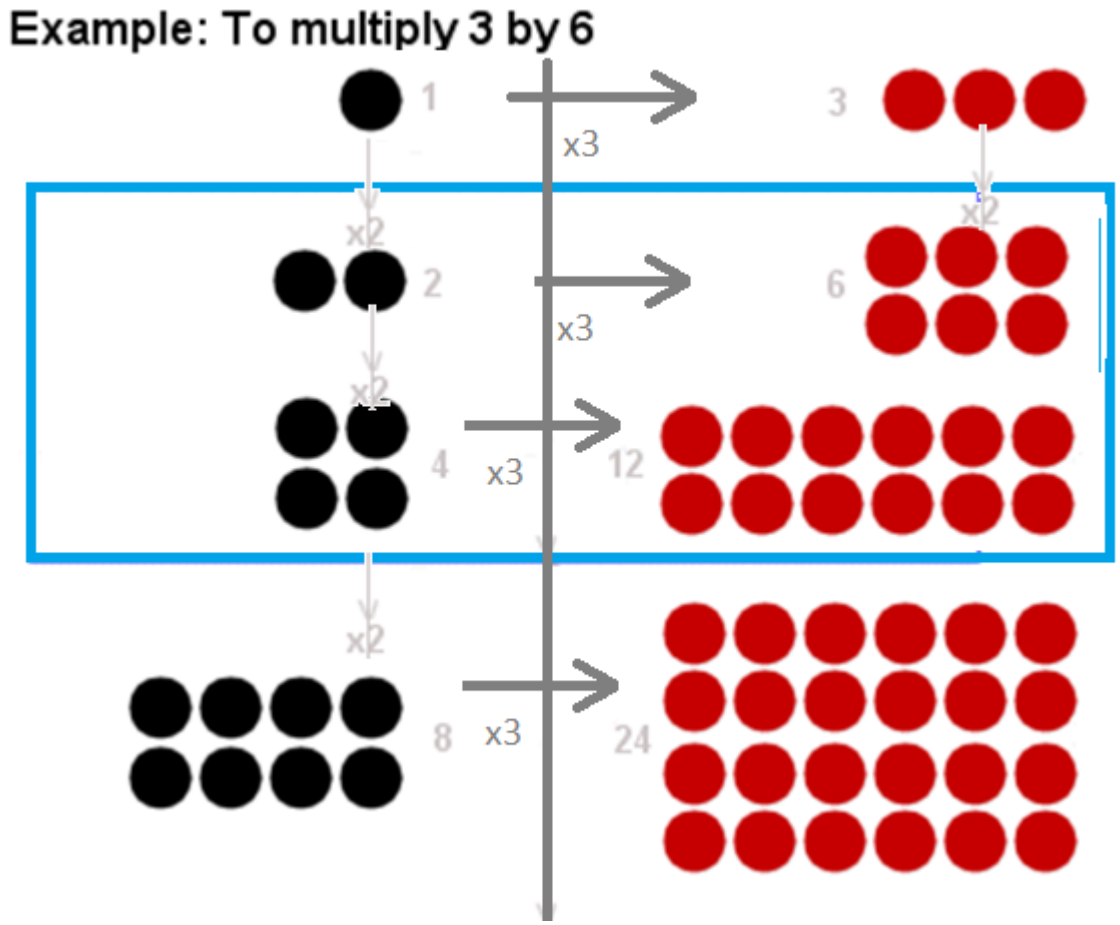


Egyptian Number System

- Please read section A and answer the problem 1a.



Egyptian Multiplication (Section B)



Write the powers of 2 on one side: 1,2,4,8,16….

Multiply each of them by the first number.

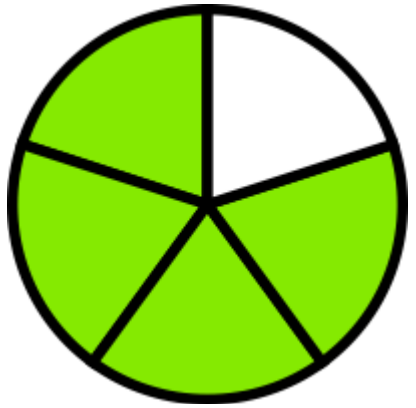
Since 2 +4 = 6, add the corresponding entries:

6 + 12 = 18

Why does this work?

Credit: http://www.storyofmathematics.com/images2/egyptian_multiplication.gif

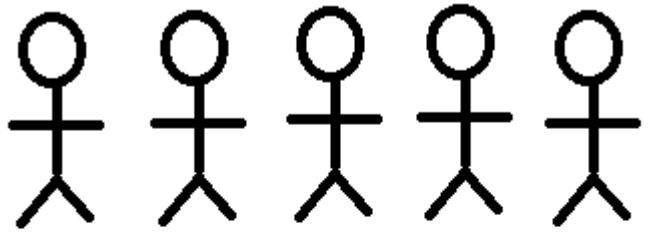
Group Discussion



Suppose 5 people order 4 whole uncut pizzas but need to share them equally. How can we share it so that

a) Everyone gets the exact same cuts of pizza

b)We want to prioritize having larger cuts



Hint: Take the first three pizzas and cut them in half and give each person a half.

Try to come up with a few solutions!



Watch video.

Egyptian Fractions

An Egyptian fraction is rewriting any fraction $\frac{a}{b}$ where $a < b$, and a, b are both positive integers as a sum of fractions where:

1. Each fraction in the expression has a numerator equal to 1, and
2. All the denominators are positive integers and are different from one another.

GREEDY ALGORITHM!

Find the largest unit fraction $\left(\frac{1}{a}\right)$ that is just smaller than your fraction. Repeat… until you end up with only unit fractions.

Not always the most elegant answer:

$$\frac{4}{17} = \frac{1}{5} + \frac{1}{29} + \frac{1}{1233} + \frac{1}{3039345} \quad OR \quad \left(\frac{4}{17} = \frac{1}{5} + \frac{1}{30} + \frac{1}{510}\right)$$

More problems in Rhind Papyrus

Linear equation with one unknown (Section D)

- Problem 31: A quantity's $\frac{2}{3}$, its $\frac{1}{2}$, its $\frac{1}{7}$ and the quantity itself added together gives 33.
  How would we solve for the quantity?

Arithmetic Progression (Section E)

E.g. 1, 3, 5, 7, 9

- Problem 40: How would we solve this problem?

Geometric Progressions



As I was going to St Ives,

I met a man with 7 wives

Every wife had seven sacks

Every sack had seven cats

Every cat had seven kits

Kits, cats, sacks, wives.

How many were going to St Ives?

Sesame Street Video.

Read the problem in section F.

How would we solve it?

Summary

- Egyptian mathematics: Rhind Papyrus
  - Numbers, Multiplication, Fractions, Linear Equations, Arithmetic and Geometric Progressions
- Tutorial next week on group project! Read the guidelines ahead of time and think about what topics might interest you.
- E-mail me with any questions or concerns: tobylai@hku.hk